14.2

18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3	
Number Entering	83	61	56	200

$H_0: p_1 = p_2 = p_3 = \frac{1}{3}$. H_a: not all same

$E = \frac{200}{3}$

$\chi^2 = \frac{(49/3)^2}{200/3} + \frac{(17/3)^2}{200/3} + \frac{(32/3)^2}{200/3} = 6.19$, $df = 2$

$\chi_{0.05,2} = 5.99 < 6.19$. reject H_0

C.I.

$\hat{p} = \frac{83}{200} = 0.415$

$0.415 \pm 1.96\sqrt{\frac{0.415 \cdot 0.585}{200}} = (0.347, 0.483)$

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.[1] In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$H_0: p_1 = p_2 = \dots = p_7 = \frac{1}{7}$. H_a: not all same

$E = \frac{200}{7}$

$\chi^2 = \frac{7}{200}\left(\frac{32^2+52^2+11^2+18^2+24^2+18^2+3^2}{49}\right) = 3.63$. $df = 6$

$\chi_{0.05,6} = 12.592 > 3.63$. non-reject H_0

14.3 **18. Fatal Accidents** Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small		Medium		Large		
Fatal	67	61.43	26	28.04	16	19.53	109
Not Fatal	128	133.57	63	60.96	46	42.47	237
	195		89		62		346

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

H_0: independent H_a: dependent

$\chi^2 = \frac{5.57^2}{61.43} + \frac{5.57^2}{133.57} + \frac{2.04^2}{28.04} + \frac{2.04^2}{60.96} + \frac{3.53^2}{19.53} + \frac{3.53^2}{42.47} \approx 1.89$.

$df = 2$. $\chi^2_{0.05,2} = 5.99 > 1.89$. non-reject H_0

DATA SET DS1403 **23. Telecommuting** As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

	Workdays at Home per Week						
Salary	Less Than One		At Least One, but Not All		All at Home		
Under $50,000	38	36.27	16	21.08	14	10.65	68
$50,000–$74,999	54	49.07	26	28.52	12	14.41	92
$75,000–$99,999	35	35.2	22	20.46	9	10.34	66
Above $100,000	33	39.47	29	22.94	12	11.59	74
	160		93		47		300

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the p-value for this test of hypothesis. Does the p-value confirm your conclusions from part a?

a. H_0: independant H_a: dependent

$\chi^2 \approx 6.45$. df = 6

$\chi^2_{0.05, 6} = 12.592 > 6.45$. non-reject H_0

b $\chi^2_{0.5, 6} = 5.35$. $\chi^2_{0.25, 6} = 7.84$

$0.25 < p\text{-value} < 0.5$.

$p\text{-value} \approx 0.375 > 0.05$. non-rejed H_0

144 DATA SET DS1406 **10. Marriage Rates** Half of adults today are married compared with the all-time high of 72% in 1960.[9] Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

	Married		
Educational Level	**Yes**	**No**	
Bachelors or higher	187	113	300
Some college	162 166	138 134	300
High school or less	149	151	300
	448	402	900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H_0: marriage rate are same . H_a: not the same

$$\chi^2 = \frac{21^2 + 4^2 + 17^2}{166} + \frac{21^2 + 4^2 + 17^2}{134} \approx 10.06$$

df = 2. $\chi_{0.01, 2} = 9.21 < 10.06$. reject H_0

There's sufficient evidence to conclude that marriage rates differ among educational levels.

Higher educational level is associated with a higher marriage rate.

DATA SET DS1407 **12. How Big Is the Household?** A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

	Type of Residence				
Family Members	**Apartment**	**Duplex**	**Single Residence**		
1	8	20	1	9.67	29
2	16	8	9	11	33
3	10	10	14	11.33	34
4 or More	6	2	16	8	24
	40	40	40		120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

H_0: distributions is same . H_a: different

$$\chi^2 = \frac{1.67^2 + 10.33^2 + 8.67^2}{9.67} + \frac{5^2 + 3^2 + 2^2}{11} + \frac{1.33^2 + 1.33^2 + 2.67^2}{11.33} + \frac{2^2 + 6^2 + 8^2}{8} \approx 36.49$$

df = 6. $\chi_{0.01, 6} = 16.812 < 36.49$. reject H_0.

Duplexes tend to have smaller households.

Single residences tend to have larger households.

Apartments are mostly two people households.